

Explaining Trends in Effective Tax Rates, 1960–2023

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Abstract

Time series of effective tax rates are the standard evidence on how progressive the tax system is, but historical comparisons conflate tax policy with changes in the distribution of income, the composition of income, and demographics. We separate these forces with a Kitagawa–Oaxaca–Blinder decomposition in which the parameters of the tax system are the coefficients and IRS microdata are the characteristics. Holding taxpayers fixed, statutory change lowered the top 0.1 percent’s effective income-plus-payroll rate by 21.6 percentage points between 1960 and 2014 and the top 1 percent’s by 16.5 points; holding the law fixed, changes in the income distribution pushed those rates back up by 14.0 and 20.9 points, respectively. The two forces nearly cancel for the top 1 percent but not at the very top, where realized rates fell. The filing-status and dependents mix never moves rates more than 3 points. Below the median the problem inverts: the non-policy forces are small—at the bottom they reinforce policy after 1979—so observed rates track the policy series: refundable credits cut the bottom quintile’s rate by 11 points, while the median’s flat rate records two policy eras canceling. Within the income effect, concentration and composition push in opposite directions, and since 1979 the shift of the very top’s income toward preferentially taxed capital gains dominates. The decomposition is robust across comparison windows and to re-anchoring the pipeline at 1984, and its policy component is invariant to the fiscal-versus-economic income concept dispute; the level, and at the very top the direction, of realized rates is sensitive to the income concept.

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1 Introduction

How progressive is the U.S. federal tax system, and how has its progressivity changed? The standard evidence is a time series of average effective tax rates by income group: the Congressional Budget Office publishes such estimates annually (Congressional Budget Office, 2026), and Piketty and Saez (2007) provide estimates back to 1960. These series anchor claims—in policy debates, media coverage, and academic research—about whether the tax code has become more or less progressive over the past several decades.

Historical comparisons of effective rates, however, conflate several distinct forces. Tax policy—rates, brackets, credits, deductions, and base definitions—changes the mapping from income to liability. But over the same decades the distribution of income became far more concentrated (Piketty and Saez, 2003), the composition of top income shifted toward preferentially taxed capital gains, and the demographic structure of the filing population (filing status, dependents) was transformed. Under any progressive rate schedule, each of these forces moves effective rates on its own, with no change in law. An observed trend in effective rates is therefore a mixture of statutory change and population change, and policy claims built on the raw trend can mislead.

This paper separates the forces with a Kitagawa–Oaxaca–Blinder decomposition (Kitagawa, 1955; Oaxaca, 1973; Blinder, 1973): the parameters of the tax system, as encoded in NBER’s TAXSIM calculator, play the role of the coefficients, and IRS Statistics of Income (SOI) microdata play the role of the characteristics. The method rests on a single cross-section of taxpayers—the 2014 SOI Public Use File—run through every year’s tax law from 1960 to 2023. Because the population is fixed, this first series moves only when the law moves; it isolates tax policy. A second series reweights the fixed sample to match each year’s actual distribution of filing status and dependents, in the spirit of DiNardo et al. (1996), adding demographic change. A third series runs each year’s actual SOI filers through that year’s law, adding the change in the level and composition of income. The gaps between adjacent series identify demographics and income effects, and the three components sum

exactly to the observed change.

Three findings emerge. First, holding taxpayers fixed, effective income-plus-payroll rates at the top fell sharply: -16.5 percentage points for the top 1 percent and -21.6 points for the top 0.1 percent between 1960 and 2014, tracing the statutory history of the Revenue Act of 1964, ERTA, TRA86, and—extending the series to 2023—the TCJA. Policy also cut rates sharply at the bottom (-11.2 points for the lowest quintile) through the creation and expansion of refundable credits, while leaving the middle roughly untouched as payroll-tax expansion offset income-tax cuts. Second, distributional change worked in the opposite direction, raising top rates under any given code: $+20.9$ points for the top 1 percent, $+14.0$ for the top 0.1 percent—mostly rising concentration pushing income into higher brackets, partly offset by the shift of top income toward preferentially taxed capital gains. For the top 1 percent the two forces nearly cancel—the realized rate changed just $+1.6$ points over six decades—but at the very top the policy cut dominates and the realized rate fell 8.5 points. Third, demographics are a minor force everywhere: never more than 3 percentage points in any group. Because the decomposition is evaluated at the 2014 population, the long-window magnitudes depend on that reference point—re-anchoring the pipeline at 1984 roughly halves the 1960–2011 policy component—though the split is anchor-robust in the post-1984 windows (Section 6).

A corollary follows: the inference problem is severe at the top and mild below it. At the top, large offsetting forces make the raw effective-rate series uninformative about policy; below the median, the non-policy forces are small—and at the bottom, since 1979, they push in the same direction as policy—so the observed path of rates for median and below-median taxpayers is, to a close approximation, the policy series itself.

Decomposing the income effect further, we find that its two ingredients are often far larger than their sum. Concentration—more income at the top—raised the top 0.1 percent’s rate by 24 points over 1960–2014, while composition—the doubling of the capital-gains share of top income, taxed at preferential rates—lowered it by 10. Since 1979 the composition force

dominates at the very top, which is why decompositions that treat “income distribution” as a single force can reach opposite conclusions depending on the window examined.

The exercise connects several literatures. Piketty and Saez (2007) document the long-run decline in top effective rates and attribute it, informally, to statutory change and to the shifting composition of top income; CBO’s annual distributional estimates (Congressional Budget Office, 2026) are the standard modern benchmark, and summary progressivity indexes (Kiefer, 1984) compress the same series into scalar measures. Closest in spirit to our approach, Burman et al. (1998) follow taxpayers across successive tax laws to separate law changes from income changes in marginal rates. A related normative literature asks what baseline should define a distributionally neutral tax change when the income distribution is itself shifting (Zelenak, 2004; Kamin, 2008). Our contribution is to make the policy-versus-population separation exact and comprehensive: the full tax calculator serves as the coefficient vector, six decades of microdata as the characteristics, and the three components sum to the observed change by construction.

Two checks support the framework. The first validates the SOI-TAXSIM measure directly: passing each year’s actual filers through that same year’s TAXSIM reproduces reported tax liability with a median R^2 of 0.996 across 1960–2011 (Appendix B), and our actual-population series moves with CBO’s income-plus-payroll rates over the 1979–2011 overlap—once CBO’s rates are renormalized to a fiscal-income denominator, the levels align as well as the trends (Appendix C). The second addresses the central measurement dispute in this literature: grossing top incomes up to the Auten and Splinter (2024) economic-income concept—and crediting the corporate tax paid on that income—leaves the policy component essentially unchanged (Section 6).

2 Empirical Approach

2.1 Three series

The first series isolates policy by holding the taxpayers fixed and letting only the law vary. It applies each year’s tax code to the 2014 SOI Public Use File—approximately 157,000 records representing 131 million tax filing units after excluding married-filing-separately and dependent returns. Incomes are deflated to each target year with the PCE price index, passed through TAXSIM (Feenberg and Coutts, 1993) under that year’s parameters—rates, brackets, exemptions, deductions, credits, payroll taxes—and the resulting liabilities are inflated back to 2014 dollars. The effective tax rate for group g under year- t law is

$$\text{ETR}_{g,t} = \frac{\sum_{i \in g} w_i (\text{tax}_{i,t}^{\text{inc}} + \text{tax}_{i,t}^{\text{payroll}})}{\sum_{i \in g} w_i Y_i}, \quad (1)$$

where w_i is filer i ’s SOI sampling weight, Y_i is filer i ’s broad income (defined below), and income groups are fixed at their 2014 percentile ranks within the filer population. We write $\text{ETR}(\beta_t, X)$ for this quantity, where β_t collects the year- t tax parameters entering the liability calculation and X collects the population’s characteristics—incomes, return items, and weights. In this first series the characteristics never change, so its movements reflect only the tax parameters: it is $\text{ETR}(\beta_t, X_{2014})$, the policy series.

The reweighted series, $\text{ETR}(\beta_t, X_{2014}^{\text{rw}})$ (“rw” for reweighted), reweights the same records so the joint distribution of filing status and dependents (twelve cells: three statuses $\times$ four dependent categories) matches each year’s actual SOI composition. Only weights change, so no new tax calculations are needed. The actual-population series, $\text{ETR}(\beta_t, X_t)$, runs each year’s actual SOI filers through that year’s law; it exists for 1960–2011, the span of the archival Public Use Files, plus the 2014 anchor—where the 2014 PUF is the actual population, so the fixed and actual series coincide there by identity. Denoting the three

series $\text{ETR}^{\text{fixed}}$, ETR^{rw} , and $\text{ETR}^{\text{actual}}$, the decomposition identity is exact:

$$\underbrace{\Delta \text{ETR}^{\text{actual}}}_{\text{Total}} = \underbrace{\Delta \text{ETR}^{\text{fixed}}}_{\text{Policy}} + \underbrace{\Delta(\text{ETR}^{\text{rw}} - \text{ETR}^{\text{fixed}})}_{\text{Demographics}} + \underbrace{\Delta(\text{ETR}^{\text{actual}} - \text{ETR}^{\text{rw}})}_{\text{Income/Inequality}}. \quad (2)$$

The tax measure is federal individual income tax plus employee and employer payroll tax; the denominator is broad income—AGI plus above-the-line deductions plus the employer half of payroll tax—following Piketty and Saez (2007), so that the measured rate is not mechanically moved by policy changes to above-the-line deductions. Appendix A details the data, the variable mappings across three generations of SOI files, the two-pass TAXSIM treatment of self-employment tax, the Net Investment Income Tax, and the Section 199A deduction, and the demographic reweighting. Appendix B reports the same-year validation: each year’s filers passed through that year’s law reproduce reported liability with a median R^2 of 0.996.

2.2 Interpretation and caveats

Three caveats govern interpretation. First, the decomposition is mechanical and partial-equilibrium. The policy component is the change in rates when 2014 taxpayers are passed through historical law—an accounting object, not an estimate of what rates would have prevailed had 1960 law persisted, since the income distribution itself responds to the tax code. Part of what the decomposition books as income and composition change—the post-TRA86 migration of business income from C corporations onto individual returns, realization timing around capital-gains rate changes—is itself a behavioral response to the statutes booked under policy, so the two components are accounting complements rather than independent causal channels, and the fixed-population policy cut at the top likely overstates the cut a behavior-inclusive counterfactual would deliver. Second, the components are defined by the labels we give the series. “Demographics” is the filing-status and dependents mix—the margins that directly enter the tax calculation—so distributional forces correlated with

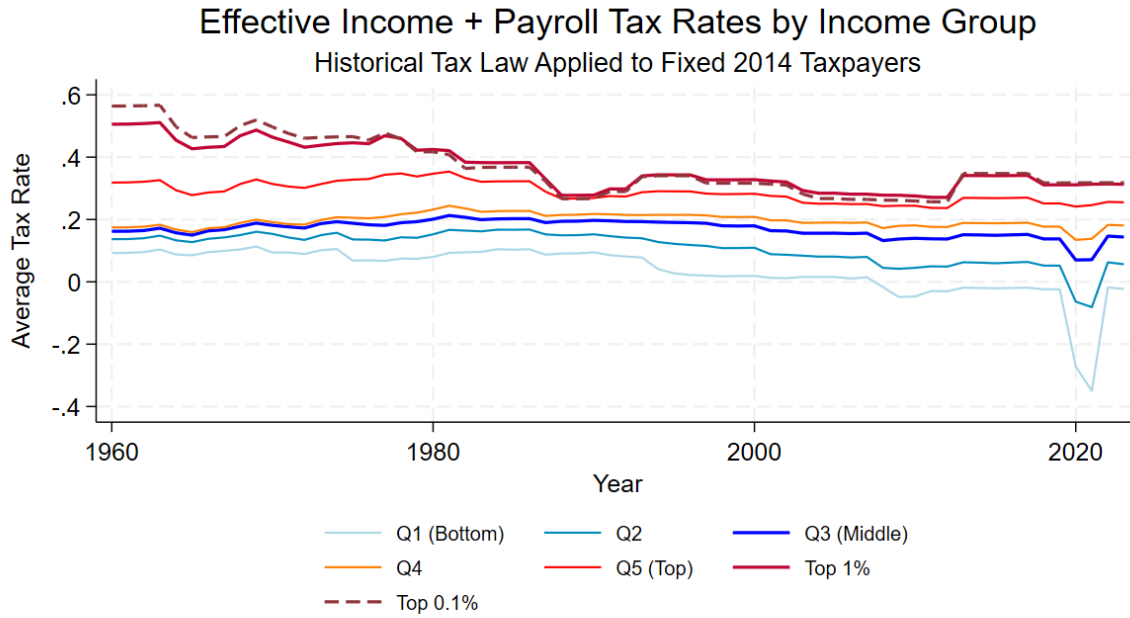
family structure (two-earner couples, assortative mating, age) land in the income residual; and because incomes are placed in each year by price deflation, the income component includes real income growth interacting with imperfectly indexed brackets, not only relative concentration. The decomposition is also sequential, with demographics evaluated at 2014 incomes. Third, Kitagawa–Oaxaca–Blinder decompositions depend on the reference point. We anchor at 2014, the most recent detailed microdata; Section 6 shows the findings are robust across comparison windows, including those confined to the best-measured decades, and to re-anchoring the entire pipeline at 1984.

3 The Policy Series: Fixed-Population Rates, 1960–2023

Applied to the fixed 2014 population, six decades of tax law cut effective rates sharply at both the top and the bottom of the distribution while leaving the middle nearly untouched. Figure 1 presents this policy series: effective rates when each year’s law, 1960–2023, is applied to the fixed population.

For the top 0.1 percent, effective rates fell from 56 percent under 1960 law to 27 percent after the Tax Reform Act of 1986, recovered to 35 percent after ATRA and the ACA taxes took effect in 2013, and fell to 32 percent under the TCJA. The 9.1 percentage point jump from 2012 to 2013 law—the payroll-tax holiday expiration, ATRA’s 39.6 percent bracket, and the ACA’s Net Investment Income Tax and Additional Medicare Tax—is the largest single-year policy change for this group in six decades. The top 1 percent follows a similar path, from 51 percent under 1960 law to roughly 34 percent by 2014 and 31 percent by 2023. Appendix Table 4 reports the rate changes at each major legislative transition.

For the middle quintile, rates fluctuate between 14 and 21 percent—setting aside the temporary 2020–2021 provisions, which briefly push the rate to 7 percent—with no trend comparable to the top: statutory rate cuts and credit expansions were approximately offset by payroll-tax growth (OASDI from a combined 6 percent on a narrow base in 1960 to 12.4



Source: 2014 SOI PUF and NBER TAXSIM35.

Figure 1: Effective Income and Payroll Tax Rates by Income Group, 1960–2023

Notes: Each year's federal income and payroll tax law is applied to the same sample of tax filing units (representing 131 million returns) drawn from the 2014 IRS Statistics of Income Public Use File. Dollar amounts are deflated to each target year using the PCE Price Index (2017 = 100). Income groups are defined by the 2014 income distribution and held fixed across all years. Recovery rebate credits for 2020–2021 are excluded to show structural rates. The effective tax rate is (federal income tax + payroll tax) / broad income, where broad income adds above-the-line deductions and the employer half of payroll tax to AGI following Piketty and Saez (2007).

Source: Authors' calculations using 2014 SOI PUF and NBER TAXSIM35.

percent on a much larger one; Medicare from zero to an uncapped 2.9 percent).

For the bottom quintile, rates fell from 9 percent under 1960 law to near zero or negative by the 2000s, driven by the EITC and refundable CTC; the deeply negative 2021 value reflects ARPA’s one-year fully refundable CTC, which we retain as a (temporary) feature of 2021 law, while stripping the 2020–2021 recovery rebates as one-time stimulus rather than tax structure.

This policy series is only half the story. Because it holds the population at its 2014 composition, it says nothing about how the changing distribution of income has moved realized rates over the same six decades—the force we isolate next by letting the population vary alongside the law.

4 The Three-Way Decomposition: Policy Against the Income Distribution

Table 1 reports the three-way decomposition of equation (2) over 1960–2014, and Figure 2 plots the three underlying series for the top 1 percent, the top 0.1 percent, and the bottom and middle quintiles, with the policy series extended through 2023. The visual contrast between the two rows of the figure previews the central result: at the top the actual-population series diverges sharply from the fixed-population series, while below the median the three series nearly coincide.

4.1 The three components

Policy. The fixed-population series traces the statutory history. At the top, the Revenue Act of 1964 (91 to 70 percent top rate), ERTA (to 50 percent), TRA86 (to 28–33 percent), and the TCJA account for a decline of 16.5 points in the fixed-population rate for the top 1 percent and 21.6 for the top 0.1 percent, with ATRA and the ACA taxes in 2013 reversing only part of the decline. At the bottom, the policy effect is equally striking: the EITC (1975;

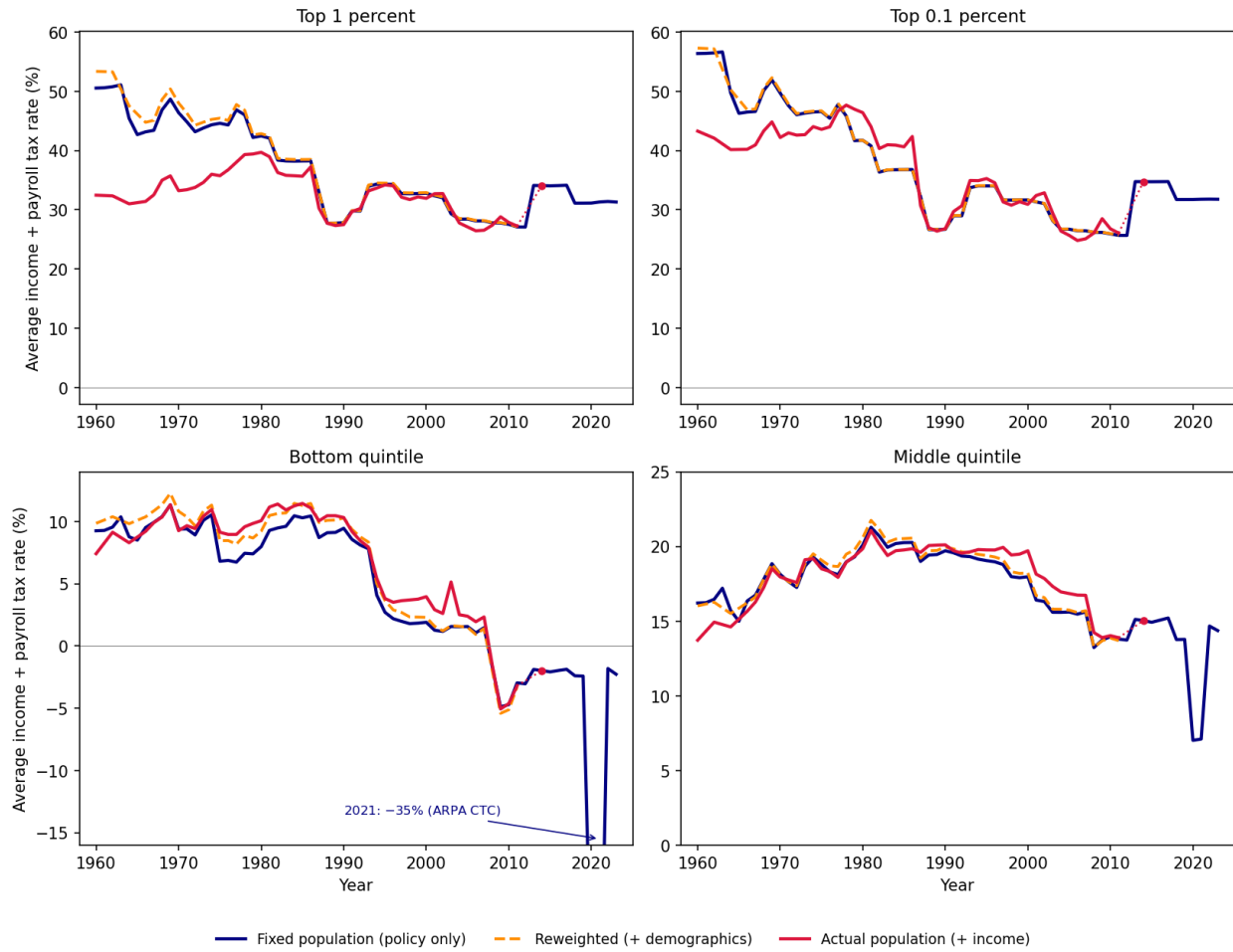


Figure 2: Three Series by Income Group: Top 1 Percent, Top 0.1 Percent, Bottom Quintile, and Middle Quintile

Notes: The fixed-population series applies each year's tax law, 1960–2023, to the 2014 SOI sample with fixed weights. The reweighted series adjusts weights to match each year's filing status $\times$ dependents distribution. The actual-population series uses each year's actual SOI filers and ends with the 2011 Public Use File; the dotted connector marks the 2014 anchor, where the three series coincide by construction. The gap between the fixed and reweighted series identifies the demographics effect; the gap between the reweighted and actual-population series identifies the income/inequality effect. The bottom-quintile panel truncates the fixed-population series' 2021 value (–35 percent, ARPA's fully refundable child tax credit) to keep the historical variation readable.

Source: Authors' calculations using SOI PUFs and NBER TAXSIM35.

Table 1: Three-Way Decomposition of Changes in Effective Tax Rates, 1960–2014

Group	Total Change	Policy	Demographics	Income/Inequality
Bottom quintile (Q1)	−9.4	−11.2	−0.6	+2.4
Second quintile (Q2)	−5.0	−7.5	−0.3	+2.8
Middle quintile (Q3)	+1.3	−1.2	+0.2	+2.3
Fourth quintile (Q4)	+3.7	+1.4	−0.6	+2.9
Top quintile (Q5)	+6.3	−4.9	−2.7	+14.0
Top 1%	+1.6	−16.5	−2.8	+20.9
Top 0.1%	−8.5	−21.6	−0.9	+14.0

Notes: Percentage point change in the effective income and payroll tax rate between 1960 and 2014 (broad income denominator). Policy is the change in the fixed-population rate (2014 filers under each year’s law), $\Delta \text{ETR}^{\text{fixed}}$ in equation (2). Demographics is the change in the gap between the reweighted and fixed series, $\Delta(\text{ETR}^{\text{rw}} - \text{ETR}^{\text{fixed}})$. Income/inequality is the change in the gap between the actual-population and reweighted series, $\Delta(\text{ETR}^{\text{actual}} - \text{ETR}^{\text{rw}})$. The three components sum to the total up to rounding. The decomposition is anchored at 2014, where the three series coincide by construction. The 1960 endpoint carries the largest measurement uncertainty: the top 0.1 percent’s actual 1960 rate has a roughly four-point band that maps one-for-one into that row’s total and income components (Appendix B).

Source: Authors’ calculations using SOI PUFs (1960–2014) and NBER TAXSIM35.

expanded 1986, 1993, 2009), the refundable CTC (2001 onward), and ARPA’s temporary expansion turn the 1960 code’s 9 percent rate on the bottom quintile into the negative rates of the modern code—an 11.2 point policy cut. In the middle, policy is a wash (−1.2 points): payroll-tax expansion offset income-tax cuts. The fourth quintile is the one group where policy raised rates on net (+1.4 points), payroll-tax growth slightly outrunning its income-tax cuts.

Demographics. Reweighting to each year’s filing-status and dependents mix moves rates by at most 2.8 percentage points in any group, and essentially zero in the middle. The direction is mildly negative at both tails—at the bottom because head-of-household and single-parent filers claim larger refundable credits than married couples at the same earnings, at the top because the married-filing-jointly share of high earners declined. In Figure 2 the reweighted series is barely distinguishable from the fixed series: whatever drove effective rates over six decades, it was not the filing-status and dependents mix. Demographics thus cannot account for the characteristics component of a two-way policy-versus-population split;

nearly all of it is income.

Income and inequality. The income component is the dominant non-policy force, and at the top it works against policy. Under any progressive schedule, more concentrated income pushes more dollars into higher brackets: applied to 1960 law, the concentrated 2014 population generates a 56 percent rate for the top 0.1 percent, while the actual, less-concentrated 1960 population faced 43 percent under the same statute. As actual concentration rose toward the 2014 benchmark, actual rates converged toward the fixed series (Figure 2), a gap-closing the decomposition attributes to income: +20.9 points for the top 1 percent, +14.0 for the top 0.1 percent (Appendix Figure 9 plots the cumulative components year by year). For the top 1 percent this almost exactly offsets the policy cut, leaving the realized rate nearly unchanged (+1.6 points) over six decades of statutory upheaval. At the very top the offset is partial and the realized rate fell 8.5 points.

At the bottom, the small positive income components (+2.4 for Q1) reflect not inequality but the changing pool of filers—a rising share of very-low-income and zero-income returns relative to the 2014 reference—a compositional drift in who files rather than a distributional force.

4.2 Below the median: where the raw series reads as policy

The bottom panels of Figure 2 show that the decomposition matters differently below the median: the three series nearly coincide, so the observed history of effective rates there is, to a first approximation, a history of policy. For the bottom quintile, the actual rate rose from 7.4 percent in 1960 to about 11 percent in the early 1980s—payroll-tax increases compounded by inflation’s erosion of then-unindexed exemptions and of the original 1975 EITC—and then fell, with only brief interruptions, to −2 percent by 2014 as the EITC expansions of 1986, 1993, and 2009 and the refundable CTC took effect. Of the −9.4 point total change over 1960–2014, policy contributes −11.2, demographics −0.6, and the filer-

pool drift just discussed +2.4; the second quintile behaves similarly (−5.0 total, of which −7.5 is policy). Measured from 1979 the alignment is even cleaner, and instructive: policy contributes −9.4 of the bottom quintile’s −11.8 point decline, while demographics (−1.3) and income (−1.2) push the same way—demographics because the shift from married to head-of-household filing raised the share of filers claiming refundable credits on the more generous single-parent schedule. So where the top’s raw series badly misstates policy, the bottom’s raw series gets policy roughly right, though it modestly overstates the policy cut since 1979: about a fifth of the observed decline reflects who files and how they file, not what the law did. These are statements about tax filers: refundable credits themselves induce filing, so the population-level policy cut differs from the filer-conditional one measured here (Appendix A).

The middle quintile’s near-flat realized rate (+1.3 points over 1960–2014) is also a story of offset, but of policy against itself rather than policy against the income distribution. Policy raised the median taxpayer’s rate by 3.1 points between 1960 and 1979, as payroll-tax expansion and unindexed brackets outweighed statutory rate cuts, and then lowered it by 4.3 points between 1979 and 2014 through bracket indexation, TRA86, the child tax credit, and successive rate cuts; demographics (+0.2) and income (+2.3) are small throughout (Appendix Table 6 reports the below-median decomposition across comparison windows). The median taxpayer’s effective rate today is about where it stood in 1960 not because the tax code stood still but because two policy eras canceled.

Why is the income effect smaller for the top 0.1 percent than the top 1 percent, when concentration rose most at the very top? Because the income component bundles two forces that push in opposite directions, and the offsetting force is strongest at the very top. We turn to that next.

5 Concentration versus Composition

The income/inequality component mixes concentration—more income accruing to top filers, pushing dollars into higher brackets—with composition—a shift in the type of income they receive, above all toward long-term capital gains taxed at preferential rates. To separate them we construct an intermediate counterfactual population. It consists of each year’s actual filers, but with each filer’s capital-gains share of AGI reset to the 2014 benchmark for their income rank—the difference absorbed into ordinary income, holding total income and rank fixed—run through the same year’s law. The gap between the actual and counterfactual populations isolates composition; the gap between the counterfactual and reweighted series isolates concentration; the two sum exactly to the income effect.

The split is stark. Over 1960–2014, concentration raised the top 0.1 percent’s rate by 24.4 points and the top 1 percent’s by 26.2—confirming the bracket-pushing mechanism at full strength—while composition subtracted 10.4 and 5.3 points respectively. The capital-gains share of top 0.1 percent AGI roughly doubled between 1979 and the 2000s, from 22 to over 45 percent, and the composition drag grows with income. Measured from 1979, composition overtakes concentration at the very top (-12.6 against $+7.3$), turning the total income effect negative—which is why the sign of the “inequality effect” in a two-way decomposition (our income/inequality component) can flip with the window examined. The capital-gains share is itself among the most tax-responsive margins of top income, so the composition component is an accounting attribution rather than a policy-invariant primitive: part of the realization behavior it records is a response to the capital-gains rates booked under policy. Appendix Table 5 reports the full split by period.

Figure 3 plots the two components for the top 0.1 percent. Each is often larger than 10 percentage points in magnitude while their sum stays within a few points of zero—two large, offsetting forces hiding inside what a coarser decomposition would call “the income distribution.” The sharp movement at 1987 is TRA86’s repeal of the capital-gains exclusion, which mechanically shrank the composition channel.

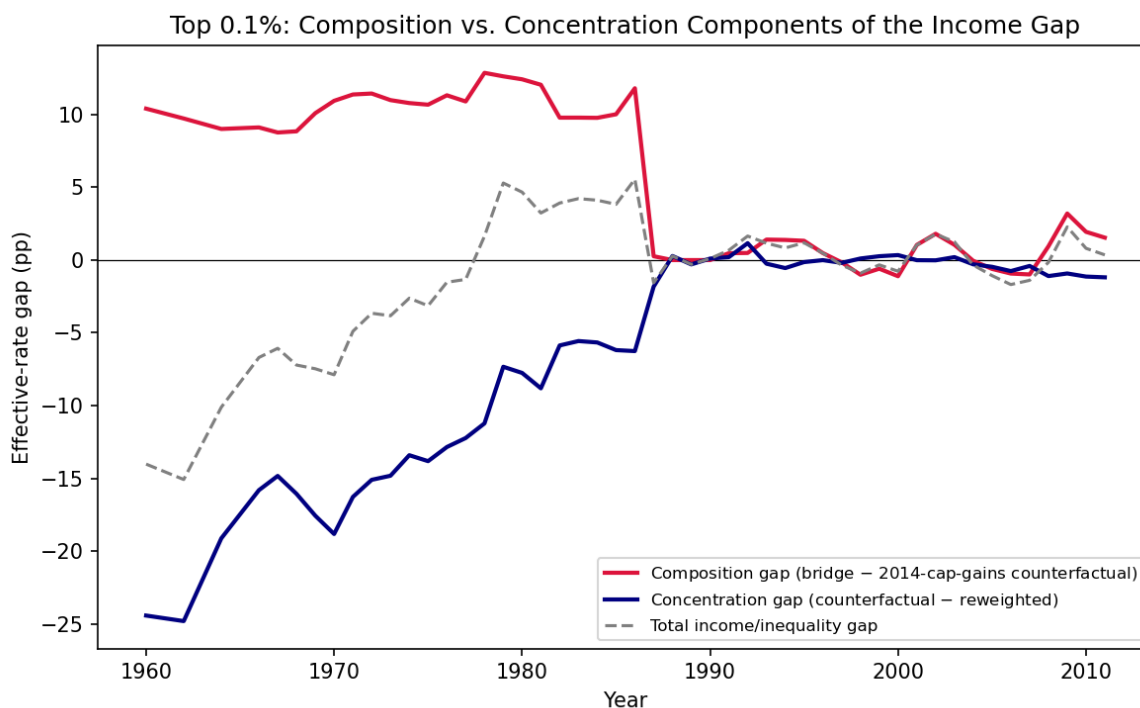


Figure 3: Top 0.1 Percent: Concentration and Composition Components of the Income Gap

Notes: Decomposition of the income/inequality gap (actual minus reweighted effective rate) into a concentration component (counterfactual minus reweighted) and a capital-gains composition component (actual minus counterfactual), where the counterfactual holds each filer’s capital-gains share of AGI at the 2014 benchmark for their income rank. The two components sum to the total gap; the series labeled “bridge” in the figure legend is the actual-population series. The sharp break at 1987 is the repeal of the 60 percent long-term capital-gains exclusion by TRA86, after which a given difference in capital-gains shares moves effective rates much less.

Source: Authors’ calculations using SOI PUFs and NBER TAXSIM35.

6 Robustness

The decomposition survives the three features of the design most likely to invite scrutiny: the 1960 starting point (the earliest and weakest-measured SOI file), the choice of the 2014 anchor population, and the fiscal-income basis of the rates. Table 2 addresses the first and third; the anchor check re-runs the entire pipeline on a 1984 base (Appendix Table 7). In every case the policy component—the paper’s central object—holds its sign and rough magnitude.

Table 2: Robustness: Comparison Windows and the Income Concept

	Top 1%			Top 0.1%		
	Total	Policy	Income/Ineq.	Total	Policy	Income/Ineq.
<i>Panel A: Alternative windows (fiscal income)</i>						
1960–2014	+1.6	−16.5	+20.9	−8.5	−21.6	+14.0
1979–2014	−5.3	−8.1	+3.2	−12.2	−6.9	−5.3
1984–2014	−1.7	−4.1	+2.7	−6.2	−2.0	−4.1
1988–2014	+6.4	+6.3	+0.0	+7.8	+8.1	−0.3
1984–2011	−8.5	−11.1	+2.9	−14.8	−11.1	−3.8
1988–2011	−0.5	−0.6	+0.1	−0.9	−1.0	+0.1
<i>Panel B: Income concept, 1960–2014</i>						
Fiscal income	+1.6	−16.5	+20.9	−8.5	−21.6	+14.0
True income + corporate tax	−5.7	−16.5	+13.5	−13.3	−21.4	+9.0

Notes: Percentage points. Total = Policy + Demographics + Income/Ineq. exactly; the demographics component (not shown) never exceeds 2.8 points in magnitude and is essentially zero in every post-1979 window, so Policy + Income/Ineq. differs from Total by at most that amount (by −2.8 points for the top 1 percent over 1960–2014). Panel A varies the comparison window; windows ending in 2011 stop at the last archival Public Use File (2011 law is pre-ATRA and includes the temporary payroll-tax reduction, so policy changes measured to 2011 are larger cuts than those measured to 2014). Panel B grosses top incomes up to the Auten and Splinter (2024) concept that includes C-corporation retained earnings and corporate tax, and adds the corporate tax, at each year’s statutory rate, to the numerator. Full window results including demographics appear in Appendix Table 6; the full income-concept analysis appears in Appendix E.

Source: Authors’ calculations; Auten and Splinter (2024).

Windows. Panel A re-runs the decomposition from alternative start years—1979 (the start of the CBO era), 1984 (a clean pre-TRA86 baseline that avoids the 1986 capital-gains realization spike), and 1988 (a clean post-TRA86 baseline)—and to an alternative 2011 endpoint that stays inside the archival microdata. The policy component simply reads the

statutory history of each window: large cuts from any pre-TRA86 start, an increase from 1988 to 2014 (ATRA and the ACA taxes), and near-zero from 1988 to 2011 (the code’s top-end stance in 2011 was still essentially the TRA86 settlement).

The income component’s behavior across windows is itself informative. From 1960 it is large and positive (concentration); from 1979 or 1984 it is small or negative at the very top (composition dominates, per Section 5); and from 1988 it is almost exactly zero. That last fact is a useful specification check: TRA86’s nearly flat 28 percent top rate made effective rates insensitive to the income distribution, so in 1988 the fixed, reweighted, and actual series nearly coincide. The income effect vanishes precisely when the rate schedule is flat—demonstrating that it is a genuine distributional force, active only under progressive rates, not an artifact of the 2014 anchor. The below-median results are stable across windows in a more direct sense: for the bottom and middle quintiles the policy component is negative in every window, and in every window from 1979 on it accounts for nearly the entire total, with the income component never exceeding 1.2 points in magnitude (Appendix Table 6).

Anchor. Kitagawa–Oaxaca–Blinder decompositions can also depend on the reference population. As a direct check, we re-run the entire pipeline—fixed series, reweighting, and decomposition—using the 1984 SOI file as the fixed population (X_{1984} in place of X_{2014}) and 1984 as the deflation base. In the well-measured windows the two anchors nearly coincide: over 1984–2011, the top 1 percent’s policy component is -10.4 points under the 1984 anchor versus -11.1 under the 2014 anchor, and the top 0.1 percent’s is -12.7 versus -11.1 , with demographics negligible and the income component keeping its sign and rough magnitude under both. Over the longest window the anchor does shift mass between policy and income—the 1984 population is far less concentrated than 2014’s, so 1960–2011 policy is -11.9 versus -23.5 for the top 1 percent—exactly the reference-point dependence the theory predicts, but the qualitative decomposition (large negative policy, negligible demographics, offsetting income) survives either choice. Appendix Table 7 reports the full comparison.

Income concept. Auten and Splinter (2024) show that fiscal income—income as reported on tax returns, the concept underlying our broad-income denominator—understates top economic income, and increasingly so going back in time: in 1960 the top 0.1 percent received less than half its true income in fiscal form, mostly because earnings were retained inside C corporations. (The size of the understatement and its implications for top-income trends are actively disputed; Piketty et al., 2018; Saez and Zucman, 2019.) If the denominator rises toward true income over time, a stable true rate could masquerade as a decline. Panel B grosses top incomes up to the Auten–Splinter concept—and, critically, adds the corporate tax paid on that income to the numerator at each year’s statutory rate, since attributing corporate income to shareholders while ignoring the 52 percent corporate rate of 1960 would treat that income as untaxed. The result: the policy component is essentially invariant (-16.5 versus -16.5 for the top 1 percent; -21.6 versus -21.4 for the top 0.1 percent), because the fixed 2014 population holds little C-corporation income to reflate. The income component shrinks (true concentration rose less than fiscal concentration, Auten and Splinter’s own point), and the total declines somewhat more (the corporate rate cut from 52 to 21 percent is itself a top-end tax reduction the fiscal series never sees). On this broader basis the realized top 0.1 percent rate is roughly flat to modestly declining rather than sharply falling—but the decomposition, our central object, does not depend on the income concept. Appendix E develops the full analysis, including the reflation factors, statutory-versus-effective corporate rates, and the re-ranking question.

Remaining caveats. The decomposition is an accounting exercise on a fixed characteristics vector, not a general-equilibrium counterfactual. The 1960 endpoint is the weakest-measured: the top 0.1 percent’s actual 1960 rate carries roughly a four-point uncertainty band that maps one-for-one into that group’s total and income components (Appendix B). And our rates cover individual income and payroll taxes only; corporate and excise taxes enter only through the Panel B exercise.

7 Conclusion

Effective tax rates are the standard measure of tax progressivity, but their movement over time mixes what the tax code did with what the population did. Separating the two with a Kitagawa–Oaxaca–Blinder decomposition—TAXSIM tax parameters as the coefficients, SOI microdata as the characteristics—shows that the two dominant forces have worked against each other at the top of the distribution for six decades. Holding taxpayers fixed, statutory change lowered top effective rates dramatically: -16.5 percentage points for the top 1 percent and -21.6 for the top 0.1 percent between 1960 and 2014, with the TCJA extending the cuts through 2023. Distributional change pushed rates back up nearly as much for the top 1 percent, whose realized rate was almost unchanged, and partially for the top 0.1 percent, whose realized rate fell. The filing-status and dependents mix never mattered more than a few points; broader family-structure change—the rise of two-earner couples, assortative mating—is absorbed into the income component by construction. And within the income force, concentration and capital-gains composition are individually far larger than their sum, with composition dominating at the very top since 1979. Below the median the decomposition validates rather than overturns the raw series: the collapse of bottom-quintile rates is nearly all policy—refundable credits, modestly reinforced by the shift toward single-parent filing—and the median’s flat rate over six decades records policy offsetting itself, payroll expansion and bracket creep before 1979 against indexation and rate cuts after.

The practical lesson for measurement is direct: a flat effective-rate series does not mean tax policy stood still, and a falling one does not mean policy alone cut taxes. For the top 1 percent, six decades of major legislation and an equally large distributional shift produced a nearly flat realized rate—a coincidence of offsetting forces, not stability of policy. The median taxpayer’s equally flat rate has a different anatomy—two policy eras canceling—and the bottom quintile’s falling rate really is policy, the one place in the distribution where the raw series can be read at face value. Assessments of progressivity, including of the TCJA and

its pending expiration, should distinguish the stance of the tax code from the distribution it happens to be applied to; the decomposition here provides a transparent way to do so.

A Data and Methods

Base sample. The fixed-population series uses the 2014 SOI Public Use File: a stratified sample of individual income tax returns with approximately 157,000 records representing 131 million tax filing units after excluding married-filing-separately and dependent returns. All series in the paper—fixed, reweighted, and actual—are computed over tax filers, with percentile groups defined within each series’ filer population; nonfilers are excluded throughout, symmetrically. This choice matters mainly for the bottom-quintile levels (filing rates were lower in early decades) and is one of the definitional wedges relative to CBO, whose universe includes nonfiling households (Appendix C). It does not affect the top groups, where filing is essentially universal. Because the PUF heavily oversamples high-income returns, sampling error is negligible for the top-group statistics the paper emphasizes.

Historical files. The actual-population series draws on archival SOI PUFs for 1960–2011 (no PUF was produced in 1961, 1963, or 1965). The files span three variable-naming eras—SOI mnemonic names (1960–1979, 1994–2002), the NBER TAXSIM–SOI crosswalk format (1980–1993), and e-codes (2003–2011)—each requiring its own mapping to TAXSIM’s input format. Three mapping subtleties matter quantitatively. First, in pre-1987 files, reported AGI is net of the capital-gains and dividend exclusions while the income components are gross, so the residual “other income” passed to TAXSIM must subtract only the amounts actually included in AGI; TAXSIM receives gross amounts and applies each law’s exclusions itself. Second, in 1994–2002 files, reported gains are pre-limitation, so the residual must use the post-loss-cap amount included in AGI. Third, for 1982–1986 the two-earner deduction is netted inside reported AGI and must be added back, and the taxpayer/spouse earned-income fields provide the wage split for married couples (an 80/20 split is used in other

pre-1987 years). Dependent ages, which govern EITC and CTC eligibility in TAXSIM, are assigned from non-income-conditional child counts where available (children-at-home 1994–2001, CTC-qualifying children 2002–2011).

Tax calculation. TAXSIM computes federal individual income tax (`fiitax`, net of all modeled credits including refundable credits) and payroll tax (`fica`, employee plus employer FICA and self-employment tax)—the $\text{tax}_{i,t}^{\text{inc}}$ and $\text{tax}_{i,t}^{\text{payroll}}$ of equation (1). For 2013–2023 law, three provisions interact in ways a single TAXSIM call cannot represent: self-employment tax (SECA), the Net Investment Income Tax, and the Section 199A deduction. We therefore use two passes. The first pass moves self-employment income into non-property income—preserving AGI while suppressing SECA—so that income tax, 199A, and NIIT are computed correctly; the second pass restores self-employment income, sets property income to the NIIT base, and computes payroll taxes correctly. Final liability combines income tax from the first pass with payroll tax from the second.

Section 199A. TAXSIM applies the 199A deduction to its business-income inputs, but the SOI does not identify qualified business income (QBI). We impute QBI from Goodman et al. (2025): claiming rates and QBI shares by AGI bin, applied stochastically to passthrough income, then split into an S-corporation component (199A without SECA) and a sole-proprietor component (199A plus SECA) using S-corporation shares of passthrough income by AGI bin from SOI Table 1.4—from 2 percent in low bins to 61 percent at the top. Before 2018 the business-income fields are folded into non-property income so that no 199A or spurious SECA arises. Two known approximations remain: the specified-service phaseout is not modeled (slightly overstating 199A for high-income service-business owners), and REIT/PTP dividends receive ordinary passive treatment.

Deflation and rebates. 2014 incomes are deflated to each law year with the PCE price index (2017 = 100) before the TAXSIM call and liabilities are inflated back, so all rates

are computed in 2014 dollars. The 2020 and 2021 recovery rebate credits are removed from liability to present structural rates; ARPA’s 2021 expanded CTC is retained as (temporary) tax law.

Reweightings. The demographic series reweights the 2014 records across twelve cells—single, married filing jointly, and head of household, by zero, one, two, and three-or-more dependents—so that each year’s cell shares match that year’s actual SOI composition. Filing status and dependents are the demographic margins that directly enter the tax calculation through rate schedules, exemptions, standard deductions, and credit eligibility. Because only weights change, the reweighted series requires no additional tax calculations.

Denominator. Broad income is AGI plus above-the-line adjustments (IRA and Keogh contributions, self-employment tax and health-insurance deductions, alimony paid, moving expenses in the years they were above-the-line, and similar items as available in each file era) plus half of each filer’s payroll tax, representing the employer share that economists treat as employee compensation. The full payroll tax is in the numerator; only the employer half enters the denominator to avoid double-counting. Broad income differs from AGI by a few percent for most filers, and the decomposition results are insensitive to using AGI instead.

B Validation of the SOI–TAXSIM Framework

The credibility of the decomposition rests on TAXSIM reproducing the tax that each year’s filers actually paid. We validate this directly with a same-year exercise: we pass each year’s SOI Public Use File through that same year’s TAXSIM calculator and compare the simulated federal income tax to the tax the SOI records as actually owed. If the mapping from SOI income and deduction items to TAXSIM inputs is correct, the two should agree closely, because TAXSIM implements the statutory code that generated the reported liability.

Tax concept. Throughout the paper the tax measure is federal income tax before the small set of credits TAXSIM cannot compute from its inputs—chiefly the foreign tax credit, the general business credit, and the prior-year minimum tax credit, none of which has a TAXSIM input field. This is not a matter of convenience but of internal consistency: the counterfactual series can only ever produce tax before these credits, because there is no way to impute, say, the foreign tax credit a 2014 filer would claim under 1960 law. We therefore hold every series to this one concept, and for the validation we add these credits back to the SOI actual tax so that both sides are measured on the same basis. The credits are small (typically one to three percent of income tax) and concentrated at the very top; excluding them consistently is cleaner than netting them on only some series. Refundable credits (the EITC and the refundable CTC) are retained on both sides, so the tax measure goes negative for filers whose refundable credits exceed their liability, exactly as TAXSIM computes.

Results. Table 3 summarizes the agreement between simulated and actual tax, weighted by SOI sampling weights. Across 1960–2011 the median coefficient of determination is 0.996, and simulated aggregate tax is within about 2.5 percent of actual in every decade. Fit is tightest from 1989 onward (median R^2 above 0.99) and remains strong back to 1960. Two features of the earliest data warrant note. First, the 1960 file is the thinnest, and its top-group agreement is the weakest in the sample (the top 0.1 percent effective rate differs by about four percentage points). Because the policy and demographic components do not depend on the 1960 actual-population rate, a four-point error in that rate maps one-for-one into the top 0.1 percent’s total change and its income/inequality component: the headline net decline of 8.5 points should be read as roughly 4 to 12 points, and its +14 point income component as roughly +10 to +18 points. The sign and qualitative conclusion—a net decline only partially offset by distributional forces—are unaffected. Section 6 reports the decomposition over shorter, better-measured windows. Second, the record-level fit for 1987 and 1988 is dragged down (to roughly 0.93) by a handful of returns—about five per year—with extreme

capital gains, for which TAXSIM applies the statutory 28 percent net-capital-gain ceiling introduced by TRA86 while the reported liability is higher; these five records account for 97–99 percent of the weighted squared error, and excluding them restores R^2 above 0.995. Crucially, this record-level sensitivity does not propagate to the group aggregates the paper actually uses: the simulated and actual effective rates agree to within about 0.2 percentage points for the middle quintile and under one percentage point for the top groups in every year, 1987 and 1988 included.

Table 3: Same-Year Validation: TAXSIM Simulated vs. Actual SOI Tax

	Years	Median R^2	Min R^2	Agg. ratio	Mean gap top 1%
1960s	7	0.999	0.961	1.016	0.8 pp
1970s	10	0.995	0.956	1.016	0.8 pp
1980s	10	0.986	0.928	1.024	0.8 pp
1990s	10	0.998	0.992	1.004	0.5 pp
2000s	10	0.996	0.991	0.989	0.5 pp
2010–11	2	0.990	0.988	0.992	0.5 pp
2014 (fixed-pop base)	1	0.994	—	1.014	1.6 pp

Notes: Each year’s SOI filers are passed through that year’s TAXSIM and compared to the SOI-reported tax, harmonized to the same before-non-modeled-credits concept (see text). R^2 , minimum R^2 , and the aggregate ratio (simulated/actual weighted total tax) are computed across the years in each decade; “Mean |gap| top 1%” is the mean absolute difference between the simulated and actual top-1-percent average income-tax rate. The final row validates the 2014 fixed-population mapping that the counterfactual series inherits: 2014 filers through 2014 law versus 2014 actual tax. Its top-group gap is larger than the historical rows because this check uses a single-pass calculation that omits the Section 199A and Net Investment Income Tax two-pass refinement present in the main pipeline, which lowers simulated tax at the top.

Source: Authors’ calculations using SOI PUFs (1960–2014) and NBER TAXSIM35.

The 2014 fixed-population mapping—the crosswalk every counterfactual series inherits—validates on the same basis at $R^2 = 0.99$, with the middle quintile matching to 0.3 percentage points. The larger top-group gap in that row reflects only that the validation check is single-pass; the full pipeline’s two-pass treatment of Section 199A and the Net Investment Income Tax reduces simulated tax at the top and closes most of the gap. Taken together, the historical and 2014 checks confirm that the SOI–TAXSIM framework reproduces actual federal income tax across the entire distribution and the full six decades, with the only material caveats confined to the 1960 endpoint and to a handful of extreme-gain records

that do not affect group-level results.

C Comparison with CBO Estimates

Our SOI-TAXSIM measure of effective rates tracks CBO’s published income-plus-payroll rates closely, which is what justifies using the series we decompose as the basis for historical comparisons of progressivity. The Congressional Budget Office publishes the most widely cited estimates of effective federal tax rates by income group, covering 1979 through 2022 (Congressional Budget Office, 2026). CBO’s estimates use administrative tax data, a microsimulation model, and supplementary survey data, and reflect the combined effect of tax policy changes and shifts in the taxpayer population; our actual-population series is an independent estimate of the same conceptual object constructed entirely from SOI microdata and TAXSIM.

CBO’s published rates are not directly comparable to ours in levels, for a mechanical reason: CBO divides taxes by income before transfers and taxes, a measure that includes substantial income invisible to an AGI-based denominator—employer-paid health insurance premiums, pre-tax retirement contributions, tax-exempt interest, social insurance benefits, and CBO’s allocation of corporate taxes to households. CBO’s published income components imply that this measure exceeds its AGI-visible components by roughly 12 percent for the top 1 percent and 20 percent for the middle quintile. Before 1987 there is an additional wedge at the top: CBO counts realized capital gains in full, while AGI of that era included only 40 percent of long-term gains. We therefore renormalize the CBO rates to a fiscal-income denominator, rescaling each group-year rate by the ratio of income before transfers and taxes to its AGI-visible components (with pre-1987 capital gains counted at the 40 percent inclusion rate), using CBO’s published component detail. The rescaling is algebraically equivalent to recomputing CBO’s taxes over an income concept comparable to ours; the figures in this section show the renormalized series, with CBO’s published rates retained for reference.

Figure 4 overlays the renormalized CBO income and payroll tax rate for the top 1 percent on our three series. The trends move together closely for the 1979–2011 overlap, and after renormalization the levels do as well: the renormalized CBO series lies within about 1 percentage point of our actual-population series for the middle quintile and within 1–3 percentage points for the top 1 percent after 1986. The small residual reflects the remaining definitional differences: CBO ranks households by market income while we rank tax units by broad income, producing different top-1-percent populations, and CBO’s universe includes nonfiling households while ours is filers only.

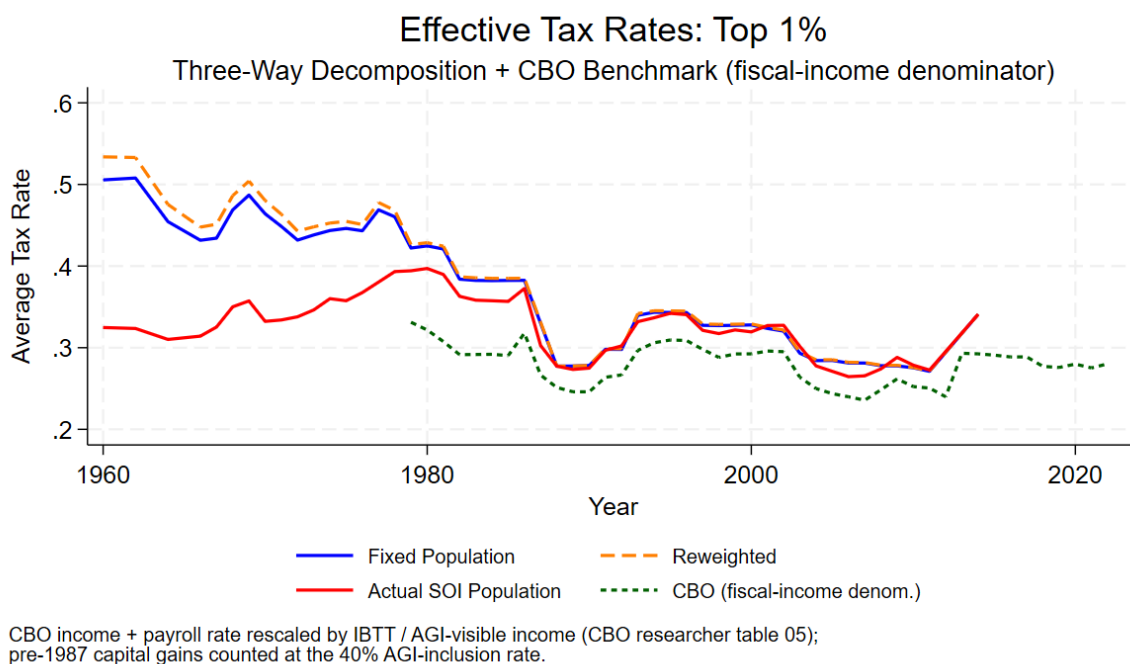


Figure 4: Top 1 Percent: Three Series with CBO Comparison, 1960–2023

Notes: The three SOI-TAXSIM series are as in Figure 2. The CBO estimate (dotted) shows the income + payroll tax rate for the top 1 percent of households ranked by market income (excluding corporate and excise taxes), renormalized to a fiscal-income denominator: each group-year rate is rescaled by the ratio of income before transfers and taxes to its AGI-visible components, with pre-1987 capital gains counted at the 40 percent AGI-inclusion rate. CBO data cover 1979–2022. The close correspondence between the renormalized CBO series and the actual-population series for the overlap period is consistent with the SOI-TAXSIM framework capturing the same first-order forces.

Source: Authors’ calculations using SOI PUFs and NBER TAXSIM35; Congressional Budget Office (2026), households ranked by market income.

For the top 1 percent (Figure 5), our fixed-population estimate shows a sustained decline from 51 percent (1960) to 31 percent (2023), reflecting the cumulative effect of income tax cuts at the top. The renormalized CBO rate tracks our actual-population series closely for the overlap period, and both sit below the fixed-population rate throughout the earlier decades of the sample (when the actual population was less concentrated than the 2014 reference) before converging in recent years as inequality rose.

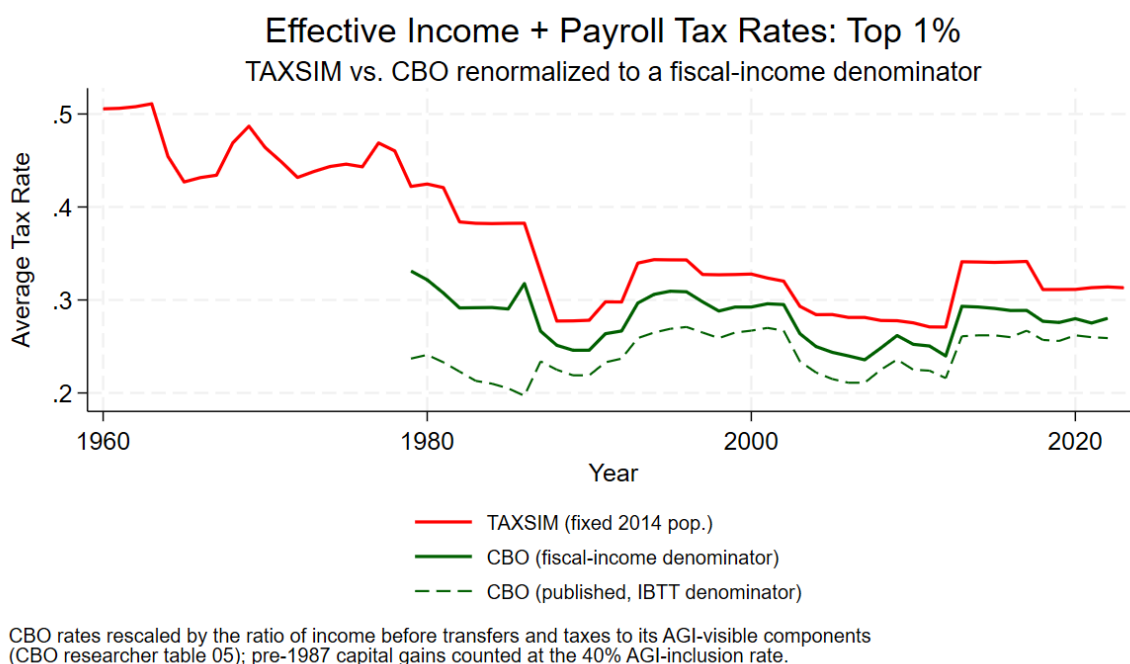


Figure 5: Effective Income and Payroll Tax Rates: Top 1 Percent, 1960–2023

Notes: The solid lines show our estimate applying each year’s tax law to the fixed 2014 SOI population and CBO’s actual effective income and payroll tax rate for the top 1 percent of households ranked by market income, renormalized to a fiscal-income denominator (see text). The dashed line shows CBO’s published rate on its income-before-transfers-and-taxes denominator. The fixed-population series declines substantially over the full period, while the CBO actual-population rate is far more stable. The closing gap between the two reflects rising income concentration: as a larger share of national income accrued to the top 1 percent, more of their income was pushed into higher brackets under any given tax code, offsetting the statutory rate cuts captured in the fixed-population series.

Source: Authors’ calculations using 2014 SOI PUF and NBER TAXSIM35; Congressional Budget Office (2026), households ranked by market income.

We extend this comparison to the top 0.1 percent using CBO’s data on tax rates for sub-groups within the top 1 percent. Figure 6 shows the result. The pattern mirrors the top

1 percent but is more pronounced: our fixed-population estimate has declined substantially, while the CBO actual rate has been more stable, reflecting the large offsetting effect of rising income concentration among the very top of the distribution.

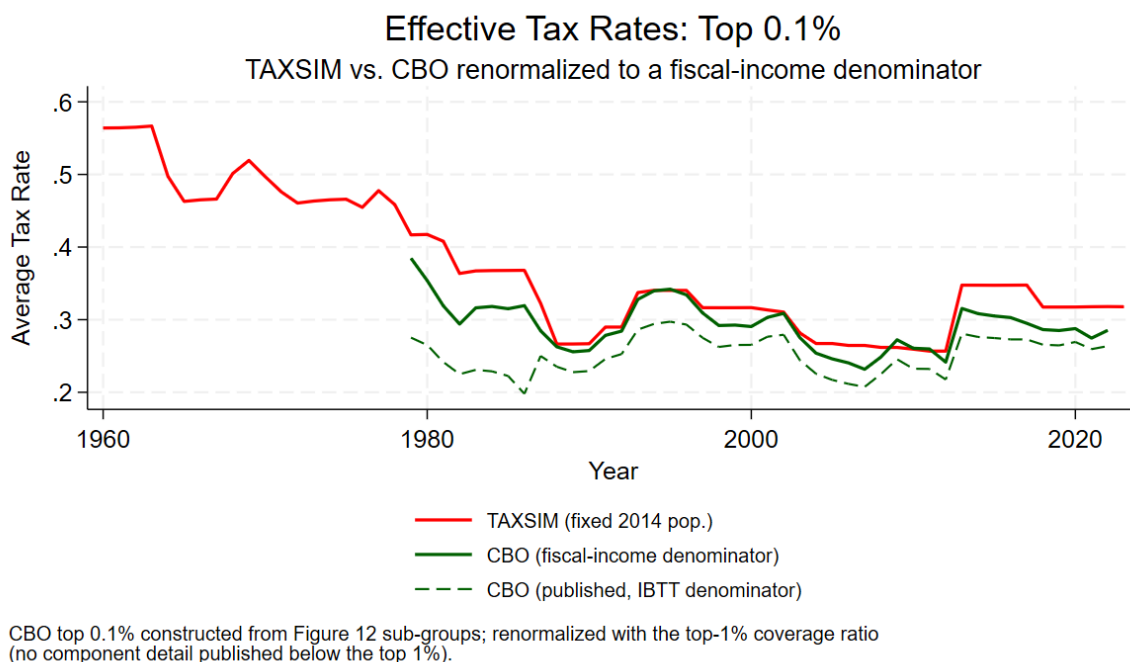
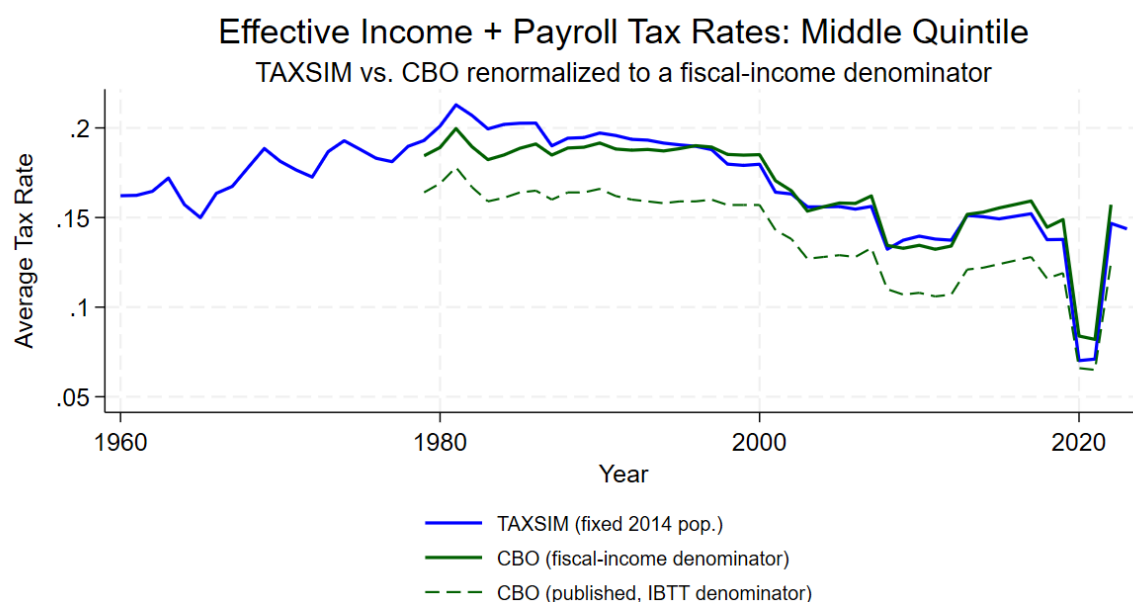


Figure 6: Effective Tax Rates: Top 0.1 Percent, 1979–2022

Notes: The solid lines show our TAXSIM estimate (income + payroll taxes, fixed 2014 population) and the renormalized CBO top 0.1 percent rate; the dashed line shows the CBO rate before renormalization. The CBO top 0.1 percent rate is constructed as an income-weighted average of CBO’s 99.9th–99.99th percentile and top 0.01 percent all-federal tax rates, with the income-plus-payroll component approximated by subtracting the corporate and excise tax rate observed for the top 1 percent. The renormalization uses the top-1-percent coverage ratio, as CBO publishes no income-component detail below the top 1 percent.

Source: Authors’ calculations using 2014 SOI PUF and NBER TAXSIM35; Congressional Budget Office (2026), Figure 12 and households ranked by market income.

Figure 7 shows this comparison for the middle quintile. Our fixed-population series declines from roughly 19 percent in 1979 to 14 percent by 2011, and the renormalized CBO series follows the same path within about a percentage point, with similar responses to major legislation—consistent with the finding that for median-income taxpayers, tax policy has been the dominant force. CBO’s published rate sits several percentage points lower throughout, entirely reflecting its broader denominator.



CBO rates rescaled by the ratio of income before transfers and taxes to its AGI-visible components (CBO researcher table 05). Source: 2014 SOI PUF/TAXSIM35 and CBO pub. 61911.

Figure 7: Effective Income and Payroll Tax Rates: Middle Quintile, 1960–2023

Notes: The solid lines show our estimate applying each year’s tax law to the fixed 2014 SOI population (the law-only, fixed-population series) and CBO’s actual effective income and payroll tax rate for the middle quintile of households ranked by market income, renormalized to a fiscal-income denominator (see text). The dashed line shows CBO’s published rate on its income-before-transfers-and-taxes denominator. CBO rates exclude corporate and excise taxes for comparability with our income-plus-payroll measure. The remaining level gap after renormalization reflects differences in the unit of analysis (tax units vs. households) and the treatment of nonfilers.

Source: Authors’ calculations using 2014 SOI PUF and NBER TAXSIM35; Congressional Budget Office (2026), households ranked by market income.

Figure 8 illustrates how CBO’s allocation of corporate income and excise taxes affects the comparison for the top 1 percent. The all-federal CBO rate is 4–6 percentage points higher than the income-plus-payroll rate, with the gap varying as statutory corporate rates and allocation methods have changed.

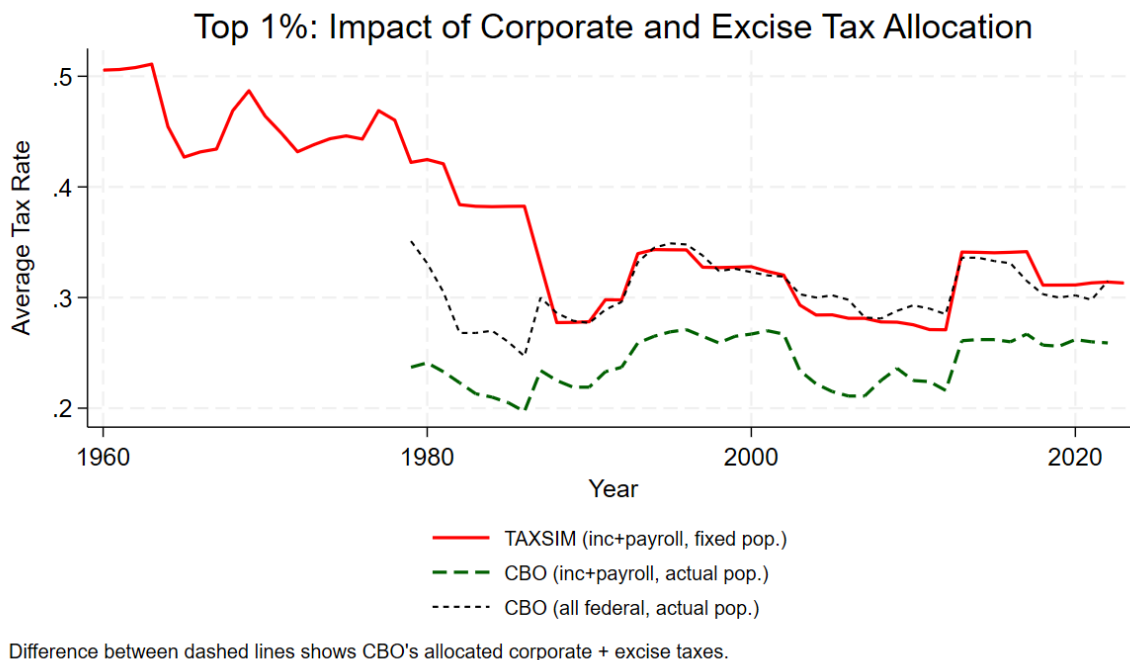


Figure 8: Top 1 Percent: Impact of Corporate and Excise Tax Allocation, 1979–2022

Notes: This figure shows three series for the top 1 percent: our TAXSIM estimate (income + payroll taxes, fixed 2014 population), the CBO income + payroll tax rate (actual population), and the CBO all-federal tax rate (actual population, including allocated corporate income and excise taxes). The gap between the two CBO series represents CBO’s allocation of corporate income taxes and excise taxes to households, which varies between 4 and 6 percentage points over the period.

Source: Authors’ calculations using 2014 SOI PUF and NBER TAXSIM35; Congressional Budget Office (2026), households ranked by market income.

D Additional Results

Legislative transitions. Table 4 reports the fixed-population rate change at each major legislative transition. Major legislation over the period includes the Revenue Acts of 1964 and

1978, ERTA (1981), TRA86, OBRA90 and OBRA93, EGTRRA (2001), JGTRRA (2003), ATRA and the ACA taxes (2013), and the TCJA (2017) (Tempalski, 2011).

Table 4: Changes in Effective Tax Rates at Key Policy Transitions

Transition	Reform	Middle Quintile	Top 1%	Top 0.1%
1963 → 1964	Revenue Act of 1964	−1.5	−5.7	−6.9
1980 → 1984	ERTA (phased in 1981–84)	+0.1	−4.3	−5.0
1986 → 1987	Tax Reform Act of 1986	−1.3	−5.3	−4.5
2000 → 2001	EGTRRA	−1.6	−0.4	−0.3
2003 → 2004	JGTRRA	−0.0	−0.9	−1.5
2012 → 2013	ATRA + ACA taxes	+1.4	+7.0	+9.1
2017 → 2018	TCJA + Section 199A	−1.4	−3.0	−3.0

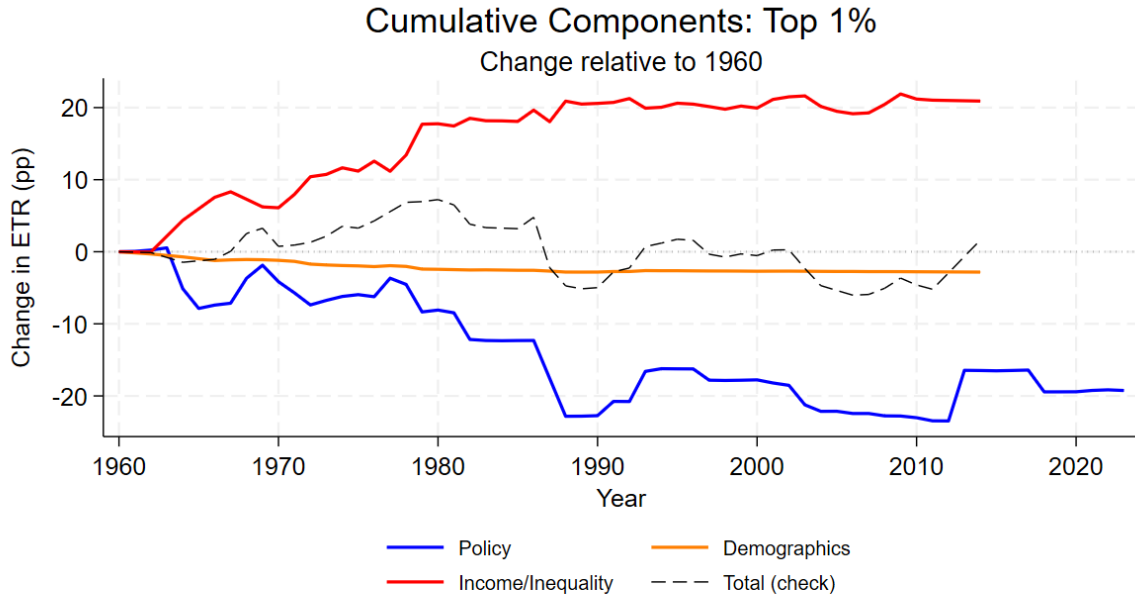
Notes: Each cell shows the percentage point change in the average effective income + payroll tax rate (broad income denominator) when the fixed 2014 population is evaluated under adjacent years’ tax law. The 2012 → 2013 increase reflects three simultaneous changes: expiration of the temporary payroll tax reduction (2 percentage points of OASDI), ATRA’s reinstatement of the 39.6 percent bracket and Pease limitation for high earners, and the ACA’s 0.9 percent Additional Medicare Tax and 3.8 percent Net Investment Income Tax. The ERTA row is cumulative over the statute’s 1981–84 phase-in; for the middle quintile, contemporaneous payroll-tax increases offset the phased-in rate cuts. Income groups are defined by the 2014 income distribution.

Source: Authors’ calculations using 2014 SOI PUF and NBER TAXSIM35.

Cumulative decomposition. Figure 9 plots the cumulative decomposition for the top 1 percent year by year, with the policy component extended through 2023.

Concentration versus composition. Table 5 reports the full concentration/composition split of Section 5 by group and period.

Window sensitivity with demographics. Table 6 reports the full three-way decomposition over all comparison windows for the bottom quintile, the middle quintile, and the top groups, including the demographics component omitted from Table 2. The below-median rows show the pattern discussed in Section 4: from 1979 on the policy component accounts for nearly the entire total, the income component never exceeds 1.2 points in magnitude, and the bottom quintile’s demographics component is uniformly negative—the head-of-household shift reinforcing the refundable-credit expansions.



Policy component (fixed population) runs through 2023;
demographics and income components end with the 2011 public-use file.

Figure 9: Cumulative Three-Way Decomposition: Top 1 Percent, Change Relative to 1960

Notes: Each line shows the cumulative change since 1960 in one component of the effective tax rate for the top 1 percent. The policy effect (change in the fixed-population rate) is persistently negative, tracing the history of statutory rate cuts, and runs through 2023; the demographics and income components end with the 2011 Public Use File. The income/inequality effect is persistently positive and large, reflecting the growing convergence of actual filers' income distribution toward the concentrated 2014 reference population.

Source: Authors' calculations using SOI PUFs and NBER TAXSIM35.

Table 5: Concentration versus Composition in the Income/Inequality Effect

Group	Period	Income/Ineq.	Concentration	Composition
Top 1%	1960–2014	+20.9	+26.2	−5.3
	1979–2014	+3.2	+9.8	−6.6
	1984–2014	+2.7	+8.5	−5.7
Top 0.1%	1960–2014	+14.0	+24.4	−10.4
	1979–2014	−5.3	+7.3	−12.6
	1984–2014	−4.1	+5.7	−9.8

Notes: The income/inequality effect from Table 1 (and its 1979 and 1984 counterparts from Table 2) split into a concentration component and a capital-gains composition component. Concentration is the change from resetting each filer's capital-gains share of AGI to the 2014 benchmark for their income rank; composition is the remainder. The two sum to the income/inequality effect. Percentage points.

Source: Authors' calculations using SOI PUFs and NBER TAXSIM35.

Table 6: Decomposition by Comparison Window (anchor fixed at 2014)

Group	Period	Total	Policy	Demographics	Income/Ineq.
Bottom quintile (Q1)	1960–2014	−9.4	−11.2	−0.6	+2.4
	1979–2014	−11.8	−9.4	−1.3	−1.2
	1984–2014	−13.2	−12.4	−1.0	+0.2
	1988–2014	−12.4	−11.1	−1.0	−0.4
	1984–2011	−14.5	−13.4	−1.3	+0.3
	1988–2011	−13.7	−12.1	−1.3	−0.3
Middle quintile (Q3)	1960–2014	+1.3	−1.2	+0.2	+2.3
	1979–2014	−4.3	−4.3	−0.5	+0.4
	1984–2014	−4.7	−5.2	−0.3	+0.8
	1988–2014	−5.0	−4.4	−0.3	−0.3
	1984–2011	−5.8	−6.4	−0.4	+1.0
	1988–2011	−6.2	−5.6	−0.4	−0.1
Top 1%	1960–2014	+1.6	−16.5	−2.8	+20.9
	1979–2014	−5.3	−8.1	−0.4	+3.2
	1984–2014	−1.7	−4.1	−0.3	+2.7
	1988–2014	+6.4	+6.3	−0.0	+0.0
	1984–2011	−8.5	−11.1	−0.2	+2.9
	1988–2011	−0.5	−0.6	+0.0	+0.1
Top 0.1%	1960–2014	−8.5	−21.6	−0.9	+14.0
	1979–2014	−12.2	−6.9	−0.0	−5.3
	1984–2014	−6.2	−2.0	−0.0	−4.1
	1988–2014	+7.8	+8.1	−0.0	−0.3
	1984–2011	−14.8	−11.1	+0.0	−3.8
	1988–2011	−0.9	−1.0	−0.0	+0.1

Notes: Three-way decomposition (percentage points) over alternative windows, all anchored at 2014. Windows ending in 2011 stop at the last archival Public Use File; 2011 law is pre-ATRA and includes the temporary payroll-tax reduction, so policy changes measured to 2011 are larger cuts than those measured to 2014. The policy effect is robust across windows; the income/inequality effect is the period-sensitive component. Effective rates use the broad income denominator.

Source: Authors' calculations using SOI PUFs and NBER TAXSIM35.

Anchor comparison. Table 7 compares the decomposition under the 2014 anchor (the paper’s baseline) and a 1984 anchor, for which the entire pipeline is re-run with the 1984 SOI file as the fixed population. The total change is identical by construction (both anchors share the same actual-population series); what the anchor can move is the split between policy and income. The 1984-anchor run uses the archival 1984 file, whose era mapping is weaker than the modern PUF’s, and performs no Section 199A imputation (immaterial here, as all comparison windows end by 2011).

Table 7: Decomposition Under Alternative Anchors (2014 vs. 1984 Fixed Population)

Group	Window	Total	Policy		Demographics		Income/Ineq.	
			2014	1984	2014	1984	2014	1984
Top 1%	1960–2011	−5.2	−23.5	−11.9	−2.8	−1.2	+21.0	+7.9
	1979–2011	−12.2	−15.1	−14.4	−0.4	−0.3	+3.3	+2.5
	1984–2011	−8.5	−11.1	−10.4	−0.2	−0.2	+2.9	+2.2
Top 0.1%	1960–2011	−17.2	−30.7	−24.2	−0.9	−0.4	+14.4	+7.4
	1979–2011	−20.9	−16.0	−20.5	+0.0	+0.0	−4.9	−0.4
	1984–2011	−14.8	−11.1	−12.7	+0.0	−0.1	−3.8	−2.1

Notes: Percentage points. Under each anchor, the fixed-population series applies each year’s law to that anchor year’s SOI filers (X_{2014} or X_{1984} , deflated on that anchor’s base) and the reweighted series adjusts that population’s weights to each year’s demographics; the actual-population series is the same in both, so the total change coincides by construction. Windows end at 2011, the last archival Public Use File. The 1984-anchor pipeline applies the same era-mapping corrections as the actual-population series (in-AGI capital gains and dividends, two-earner deduction, taxpayer/spouse wage split) and passes gross Social Security benefits so that each law year’s inclusion rules apply.

Source: Authors’ calculations using SOI PUFs and NBER TAXSIM35.

E Auten–Splinter Income-Concept Robustness

This appendix develops the income-concept robustness summarized in Section 6. The concern of Auten and Splinter (2024) is that fiscal income—our denominator—is a rising share of true economic income at the top, so that a stable true effective rate can appear as a decline. Their online Table C1 reports top-fractile income under fiscal income and under

a sequence of progressively broader concepts, ending in pre-tax national income. For each concept we form the ratio of that concept’s top-group income to fiscal income, by year and fractile; this ratio is the factor by which we reflate the denominator.

Reflation factors. Table 8 reports the factors for the top 0.1 percent. They are large early and shrink over time, with the sharpest compression at TRA86: in 1960 the top 0.1 percent received under half of its true income in fiscal form; by 2014 the fiscal and broader concepts nearly coincide, because business income that once sat inside C corporations now flows through partnerships and S corporations onto individual returns.

Table 8: Reflation Factors for the Top 0.1 Percent (broader income $\div$ fiscal income)

Year	+ C-corp retained	+ C-corp taxes	Pre-tax national income
1960	1.83	2.19	2.48
1979	1.72	1.88	2.24
1984	1.43	1.48	1.78
1988	1.09	1.13	1.29
2000	1.00	1.03	1.19
2014	1.02	1.05	1.25

Notes: Ratio of top-0.1-percent income under each concept to fiscal income, from Auten and Splinter (2024) online Table C1. Concepts are cumulative.

The corporate tax. Grossing up the denominator to include corporate retained earnings while leaving only individual and payroll taxes in the numerator would treat that corporate income as untaxed. It was not: the statutory top corporate rate was 52 percent in 1960 and remained at or above 46 percent through 1986, before falling to 34, then 35, then 21 percent. We therefore add the corporate tax to the numerator at the statutory corporate rate applied to the attributed corporate income. Doing so keeps the early-year top rate high—the corporate income of the 1960 top 0.1 percent was taxed heavily—so that the realized effective rate tracks the fiscal series rather than collapsing. The naive denominator-only reflation, which sets the corporate rate to zero, lowers the 1960 top-0.1-percent rate from 43

to 20 percent and manufactures an apparent rise over the period; this is the spurious dotted series in Figure 10.

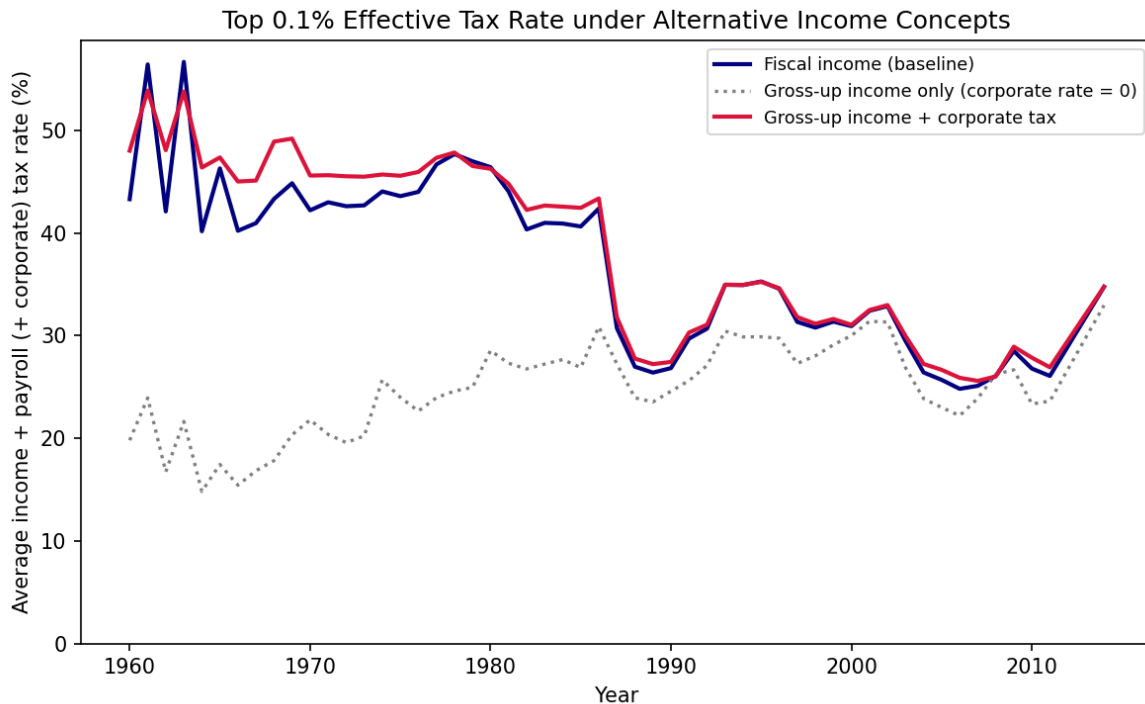


Figure 10: Top 0.1 Percent Effective Tax Rate under Alternative Income Concepts, 1960–2014

Notes: The average income and payroll tax rate for the top 0.1 percent under three treatments: fiscal income (our baseline); income grossed up to include C-corporation retained earnings and corporate tax, with the corporate tax added to the numerator at the statutory rate; and the same gross-up with the corporate tax omitted (an implicit zero corporate rate). The consistent gross-up tracks the fiscal baseline; the corporate-rate-zero version diverges.

Source: Authors’ calculations using SOI PUFs and NBER TAXSIM35; income concepts from Auten and Splinter (2024); statutory corporate rates from published rate schedules.

Decomposition across concepts and periods. Table 9 reports the three-way decomposition on fiscal income and on the corporate-adjusted concept, over two start years. The policy component is essentially invariant to the income concept in every row. The income/inequality component is uniformly smaller on the broader concept, reflecting that true concentration rose less than fiscal concentration. The total change is more negative on

the broader concept, because the corporate rate cut from 52 to 21 percent is itself a large reduction in top-end taxation that the fiscal-income series does not see.

Table 9: Three-Way Decomposition: Fiscal versus Corporate-Adjusted Income

Group	Period & basis	Total	Policy	Demographics	Income/Ineq.
Top 1%	1960–2014, fiscal	+1.6	−16.5	−2.8	+20.9
	1960–2014, true + corp.	−5.7	−16.5	−2.7	+13.5
	1984–2014, fiscal	−1.7	−4.1	−0.3	+2.7
	1984–2014, true + corp.	−4.1	−4.4	−0.3	+0.6
Top 0.1%	1960–2014, fiscal	−8.5	−21.6	−0.9	+14.0
	1960–2014, true + corp.	−13.3	−21.4	−0.9	+9.0
	1984–2014, fiscal	−6.2	−2.0	−0.0	−4.1
	1984–2014, true + corp.	−7.8	−2.5	−0.0	−5.3

Notes: Percentage points. “True + corp.” grosses income up to include C-corporation retained earnings and corporate tax and adds the corporate tax at the statutory rate. The fixed and reweighted series use the 2014 population’s corporate income; the actual-population series uses each year’s. Policy is essentially unchanged across bases.

Source: Authors’ calculations; Auten and Splinter (2024).

Statutory versus effective corporate rates. We apply statutory corporate rates for transparency and consistency with our treatment of the individual code. Using instead the lower effective corporate rates implied by Auten and Splinter (2024)’s own attribution (which reflect credits, foreign income, and bracket structure) raises the early-year corporate-adjusted rate less, leaving the realized top-0.1-percent rate roughly flat over 1960–2014 rather than declining. The policy component is unaffected by this choice, since it depends on the 2014 population, which holds little corporate income.

Denominator reflation versus re-ranking. This exercise reflatates each top group’s denominator but retains our fiscal-income ranking of who belongs in the group. A fuller treatment would re-rank tax units on true income. Because the fiscal-income top and the true-income top are largely the same filers—both are the owners of corporate capital—re-ranking would change the composition of the groups only at the margin, and would not alter the total-rate conclusion, which is driven by the level of the denominator. Its main effect

would be to refine the income/inequality attribution. A full re-ranking requires imputing each missing income component—dominantly C-corporation retained earnings and the corporate tax on them—to individual filers in proportion to observed capital income, scaled to the national-accounts aggregates; we leave this to future work.

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